NYSA CORE WhatsApp campaigns and AI conversations

Requirements and solution design for business approval

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Status For approval | Not yet an approved baseline

Decision requested

Approve the business requirements, controlled campaign workflow and phased delivery described here. We will first establish protected 360dialog sandbox connectivity, then deliver bulk campaigns, followed by AI-assisted responses and accepted Agent handover. The requested user-release target is 8 November 2026; release depends on the acceptance criteria and external account readiness, not the date alone.

This document consolidates the discussion into one approval candidate. Explicitly agreed requirements are labelled Confirmed. Detailed implementation choices and numerical defaults are Proposed for approval. Items in the decision register require resolution before the affected feature can be baselined or enabled.

What approval does and does not authorise

Business approval fixes the scope and decision rules for implementation and CRM Test acceptance. It does not approve production activation, number migration, deletion of REM assets, provider contract acceptance, campaign expenditure or sending a particular live campaign. Those actions retain their separate approvals.

Control	Position
Existing application	CRM Test DEV254; production RC1. Latest observed evidence is dated 31 October and 1 November 2026.
Source rollback baseline	DEV254 GitHub commit 42b25cb4d84d93a33336a2c49149759d4b2ca9b6; database migration 159.
New development	Branch codex/dev255-whatsapp-sandbox created. No WhatsApp feature implementation or sandbox-key setup is represented as complete.
Credentials	No sandbox key has been received or stored. Only protected entry will be used.
Production and REM	Remain unchanged and on hold for this work. No number cutover is authorised by this document.

1 Scope and agreed requirements

We will use 360dialog as the WhatsApp transport and CORE as the authority for recipients, consent evidence, campaigns, conversations, Leads, inventory and assignment. The first live delivery must support controlled bulk messaging and a separately enabled AI follow-through capability.

ID	Requirement	Basis
WA01	Support bulk WhatsApp campaigns, then AI conversations and follow-through to Agent handover.	Confirmed
WA02	Import new prospects as well as use permitted existing CORE Customers and Leads. Name and phone are mandatory; other attributes are optional.	Confirmed
WA03	Agents prepare campaigns. Manager or Director approves the campaign template and actual messaging before sending.	Confirmed
WA04	Approved follow-through may share prospectuses, brochures, floor plans, marketing material and recorded pricing when requested.	Confirmed
WA05	CORE must verify Meta template status automatically. A manual approval checkbox cannot authorise sending.	Confirmed
WA06	New templates may be created afresh. Use of REM templates is not required.	Confirmed
WA07	Sandbox recipient is the privately recorded phone ending 6469. The live business number ending 8022 remains separate.	Confirmed
WA08	Retain existing CORE workflows, permissions and authoritative ownership unless an explicit change is approved.	Required boundary

Included in the design

Protected setup, recipient import and reconciliation, grouping, permission checks, template catalogue, internal approvals, queued dispatch, callbacks, conversation history, approved material sharing, follow-up scheduling, qualification, meeting requests, Agent acceptance, reporting, recovery and release controls.

Excluded from this approval candidate

REM historical-chat import or cancellation; number migration itself; voice calling; autonomous negotiation, discounts, reservations, contracts or eligibility decisions; unrestricted web browsing by the AI; automatic creation of arbitrary new business lines or teams; unrelated CRM corrections. Sandbox messages are tests, not real bulk campaigns.

2 Current capability and delivery gaps

These findings describe inspected code and deployment evidence, not a claim that WhatsApp connectivity has passed. A built adapter, a saved key and a provider receipt are distinct milestones.

Area	Existing position	Required next work
Inbound route	Test route exists but returned 503 WHATSAPP_INBOX_DISABLED. The corresponding production route returned 404.	Separate sandbox receiver; configure authentication and prove provider callbacks.
Outbound adapters	Text and basic template adapters exist. They target the live 360dialog address and restrict sending to local or Test environments.	Sandbox host support, protected configuration, bounded test sends and live release controls.
New prospects	Current reply processing correlates to prior outreach; unmatched messages are held for review.	Conversation identity and governed Lead creation for new interested prospects.
Templates	Basic named-template sending exists.	Provider catalogue sync, status freshness, version matching and send-time checks.
Bulk campaigns	The complete approved audience and dispatch workflow has not been demonstrated or deployed.	Import, audience snapshot, approval, queue, suppression and reconciliation.
AI and handover	Preliminary follow-up and chat components exist behind disabled controls.	Integrate approved knowledge, manual takeover, bounded AI tools, meeting and acceptance workflow.
Tests	Connector tests use simulated providers; existing CRM regression is green.	Real sandbox and limited live-channel tests, including media and custom templates.

Baseline and scope of change

Implementation will extend isolated WhatsApp modules and reuse established CORE services where their contracts match. It must not repurpose a partial old workflow without validating its consent, ownership and retry semantics. Database tables and migration numbers below are design proposals; none is claimed to have been applied.

The existing Test webhook path is /api/providers/lead-whatsapp/events. It is not a ready sandbox endpoint. The sandbox documentation has a different event shape, including examples without live account and phone metadata; the live handler cannot simply be enabled with a sandbox key.

3 End to end business journey

Step	Business event	CORE outcome
1	Agent uploads an audience or selects permitted CORE records.	Validated batch, duplicate links, permission status and count reconciliation.
2	Agent selects a segment and campaign purpose.	Recipient preview with explicit exclusions and optional manual removals.
3	Agent prepares or reuses an internally approved template.	Manager or Director approval of wording; authorised Admin submits to Meta through 360dialog.
4	Provider reports the template usable.	Verified catalogue record tied to account, language and content version.
5	Agent prepares personalised campaign and follow-through package.	Fixed recipient snapshot, exact rendered messages, materials, schedule and budget.
6	Manager or Director approves the campaign revision.	Immutable approval evidence; queue becomes eligible for the approved schedule.
7	CORE dispatches to eligible recipients.	Per-recipient attempt ledger, provider receipt and subsequent status updates.
8	Prospect replies or asks for materials and pricing.	One conversation, matched identity, approved answer or human-review task.
9	Prospect expresses a requirement or meeting interest.	Existing Lead reused appropriately or governed new Lead created once; requirement evidence retained.
10	Meeting or callback is arranged.	Verified slot reserved where connected, otherwise a preference awaiting confirmation.
11	Responsible Agent accepts handover.	Owner, next action and due time recorded; AI sales follow-up pauses.
12	Agent completes the next action.	Existing Opportunity, viewing and deal workflows continue unchanged.

A reply such as STOP or a wrong-number report is not a success case. It immediately suppresses the affected marketing and cancels pending follow-up. A greeting alone must not be treated as a qualified Lead or booked meeting.

Success is reported at separate stages: message delivered, engaged prospect, enquiry created, meeting requested, meeting confirmed and Agent handover accepted. None of these states substitutes for another.

4 Recipient upload and identity

IMP01 Accept Excel XLSX and CSV through a column-mapping preview. Require name and phone for each imported recipient; preserve the original row number and supplied values as evidence.

IMP02 Support optional email, transaction interest, business line, area, budget and currency, property type, minimum and maximum bedrooms including fractions, timeframe, language, source, tags and permission evidence. Missing values remain not provided.

IMP03 Normalise phone numbers to international form using an explicitly selected country only where needed. Do not silently assume a country or convert an ambiguous number. Flag invalid or ambiguous rows for correction.

IMP04 Deduplicate by normalised channel identity within the batch and across the audience. Match existing CORE contacts without overwriting their identity, owner, requirement or consent evidence. Multiple plausible matches require review.

IMP05 Retain imported non-customers as campaign prospects. Import alone does not create a Customer, Lead, Opportunity or assignment. On a genuine enquiry, use the approved promotion service with an idempotent conversation-to-Lead link.

IMP06 Show row-level reasons and a downloadable error file. An import must never silently drop rows or claim full success after partial acceptance. Escape spreadsheet formula content in exported reports and scan uploaded files.

Count reconciliation

Every parsed data row receives exactly one disposition: accepted new, linked existing, duplicate, invalid or awaiting review. Their sum must equal the parsed row count. Blank rows and header treatment are reported separately. Messaging eligibility is a separate count over unique resolved recipients; exclusion reasons must not inflate or reduce the unique total through double counting.

Example disposition	Rows
Accepted new prospect	150
Linked existing identity	20
Duplicate row	15
Invalid row	10
Awaiting review	6
Total parsed rows	201

In this synthetic example, 170 unique recipients are ready for eligibility evaluation. If 130 pass, the remaining 40 must be explained. The application must not report that all 201 were uploaded as messageable recipients.

5 Audience grouping and permission

AUD01 Provide saved segments combining transaction interest, business line, project, property type, area, budget, fractional bedroom range, timeframe, relationship stage, source, tags, language and responsible team or Agent. Preserve existing access scope.

AUD02 Allow imported batches and existing CORE contacts to contribute to a segment without double sending to the same normalised phone. Shared or conflicting identities require review rather than disclosure of another person's information.

AUD03 Show filter definitions, matching unique count, last evaluation time and exclusions. Include not provided as an explicit filter option. Uploaded claims about requirements remain distinct from verified CORE requirements.

AUD04 Keep saved segments dynamic. Freeze an approved campaign's recipient membership and rendered variables at submission. New matches cannot be silently added later; new recipients require a new campaign revision and approval.

CON01 Require recorded WhatsApp marketing permission and its scope, source, capture time and evidence before campaign dispatch. Name, phone, an imported list, prior email contact or Manager approval is not itself permission.

CON02 Allow batch evidence only where it genuinely covers each identified recipient. Store missing, disputed, withdrawn and expired permission separately. No default consent checkbox may be preselected on import.

CON03 Recheck suppression, permission, recipient identity and campaign eligibility immediately before each send. The latest restrictive decision wins. Opt-outs apply across batches, duplicate prospect rows and linked CORE records.

Example segment

A campaign might target apartment buyers interested in Dubai Marina, with a budget of AED 2.5-3.5 million and a requirement of 2-2.5 bedrooms. A separate campaign could target an exhibition batch with unknown requirements. Both still need the same eligibility checks; incomplete profile details must not become inferred qualifications.

Frequency and jurisdiction

Contact frequency, allowed sending hours, geography and consent retention are configurable policy controls that require the business decisions in section 20. Recipient local time should be used when known; unknown timezones must follow the approved fallback, not a guessed location. Applicable marketing rules require separate business review.

6 Template lifecycle and Meta verification

There are three distinct approvals: NYSA approval of reusable template wording, Meta approval of the WhatsApp template, and NYSA approval of the particular campaign. Meta approval does not certify a property claim, a recipient list or a pricing promise. [S1]

TMP01 Agent drafts the template purpose, language, category, variables and example values in CORE. Manager or Director approves the internal wording. An authorised Admin then submits the approved version through the 360dialog Hub for the first delivery; direct submission from CORE is a later extension unless separately included.

TMP02 Synchronise the connected account's template catalogue through the authorised provider API. Record provider template ID, name, language, category, component structure, content fingerprint, status, observation time and account/channel binding.

TMP03 Permit campaign use only when the exact template version is provider-approved and usable. Unknown, stale, pending, rejected, paused, disabled or deleted templates block dispatch. A manually entered approval cannot override this control.

TMP04 Check the provider before a campaign starts and during dispatch according to a bounded freshness policy. Process authenticated status notifications where supported and periodically reconcile through the API. Proposed freshness is 5 minutes; final limits are a decision item.

TMP05 Pause queued sends when status, category, content, language or variable structure changes. Content changes require a new NYSA approval and valid Meta status. A safe resume must not repeat accepted or unresolved attempts.

TMP06 Validate all recipient-specific variables before submission. Manager sees the actual rendered messages and differences, not merely one generic template sample. Missing required values are exclusions or explicit corrections; they cannot be invented.

Template content supported at release

The first bulk release should support approved text plus document or image headers and supported URL or quick-reply buttons where required by the agreed campaigns. Each component requires a validated provider mapping and live-channel acceptance test. The current basic adapter does not establish this support. AI-requested brochure and floor-plan delivery is a separate session-message capability, subject to the messaging window.

No dependency on REM templates

NYSA may create fresh templates. If the provider migration copies eligible old templates, they remain unselected unless separately approved for CORE use. Template deletion and number migration remain separate operations. [S6]

7 Campaign approval and immutable versions

APR01 Agents can create and revise drafts, preview recipients and messages, attach approved material and submit for review. Only an authorised Manager within the governed team scope or a Director within company scope can approve sending.

APR02 Approval binds the campaign revision, sender channel, template version, full rendered message manifest, recipient snapshot, material versions, pricing sources, schedule, expiry, maximum spend and follow-through policy.

APR03 Review must expose recipient totals and exclusions, message variants, all attached files, estimated cost with pricing basis, and unresolved warnings. The reviewer can inspect every rendered recipient message without exporting unrestricted personal data.

APR04 Changes to content, variables, eligible recipient additions, sender, material versions, schedule beyond the approved bounds, budget or AI policy invalidate approval. A reduced set caused by new opt-outs may be skipped without reapproval, but additions are never allowed.

APR05 A rejected or returned campaign retains its decision and reason. Resubmission creates a new revision. Approval expiration, approver-role change and withdrawn approval are checked before launch and each queued send.

APR06 Campaign launch occurs only from the approved revision at the approved time. Proposed default is scheduled server dispatch after approval, with an immediate pause control. Allowing an Agent to launch manually after approval is a separate decision; approval never permits editing the approved package.

Follow through is included in approval

Manager or Director approves the assistant's permitted topics, document catalogue, pricing sources, qualification questions, follow-up templates and timing, maximum attempts, handover conditions and AI limits. Routine replies within this package do not need a new approval for each sentence. An unsupported claim or deviation must be escalated.

Separation of duties

Proposed control: the preparer cannot approve their own campaign, including a Manager or Director who creates one. A second eligible reviewer is required. This is an approval decision because the conversation confirmed Manager/Director approval but did not decide self-approval. Any exception must be explicit and auditable.

8 Dispatch states and recovery

Object	States	Rule
Campaign	Draft, awaiting template, awaiting approval, approved, scheduled, sending, paused, completed, cancelled	Completed means all recipients have a terminal dispatch disposition, not that all messages were delivered.
Recipient send	Queued, excluded, reserved, submitting, accepted, rejected, outcome unknown, cancelled	Accepted needs a provider message ID. A timeout after submission is outcome unknown.
Delivery evidence	Unknown, sent, delivered, read, failed	Separate from dispatch and business outcome. Absent read evidence does not mean unread.
Follow-up job	Scheduled, leased, submitted, cancelled, review required	Recheck conversation ownership, opt-out, window and policy generation immediately before sending.

SND01 Use a durable per-recipient queue and unique campaign-revision-recipient-action key. Worker leases, retries and concurrent approval actions must not create duplicate submissions.

SND02 Reserve budget and quota atomically before contacting the provider. Rate-limit for the connected account, provider responses and recipient frequency policy. Estimates must be labelled estimates; no hardcoded price or assumed delivery guarantee is acceptable.

SND03 Never blindly retry an outcome-unknown submission. Reconcile using provider IDs, callbacks or an operator review. A provider that cannot establish the outcome requires a visible unresolved state, not a fabricated failure or automatic resend.

SND04 Retry only classified safe transient failures, with bounded backoff. Invalid recipients, permissions and template failures are not retryable without correction. Preserve provider error codes but redact private provider responses.

SND05 Pause must prevent new submissions and cancel unsubmitted work as appropriate. Already submitted messages cannot be recalled; the UI must say so. Restart must resume from durable state without resetting counts or budget.

Recipient-count reconciliation must explain every member of the approved snapshot as queued/in-flight, accepted, rejected, excluded, cancelled or unresolved, with mutually exclusive dispatch states. Delivery/read counts are a second set over accepted attempts and must not be added to the dispatch total.

9 Incoming messages and Lead promotion

IN01 Acknowledge a valid provider event promptly after durable receipt, then process asynchronously. Authenticate the provider connection, bind the channel/account, enforce body bounds and deduplicate message and callback IDs. [S3]

IN02 Create or reuse one conversation identity for the connected sender and prospect. Do not depend solely on a reply-to reference: unsolicited enquiries and replies without context must still reach a visible inbox or review queue.

IN03 Correlate a reply to its original campaign when evidence permits. For overlapping campaigns or ambiguous shared phones, preserve the ambiguity and route for review rather than attributing it to the most convenient campaign.

IN04 Match existing CORE identity by normalised phone and other authorised evidence. Do not reveal existing private details to the sender merely because a phone matches. Conflicts and reassigned numbers require human review.

IN05 Promote a new prospect to a Lead once the message expresses an actual enquiry or requirement, under a defined, idempotent promotion rule. Record source campaign, original imported prospect and evidence. Reuse an appropriate active Lead where possible; a distinct later requirement may need a new Lead under existing CORE rules.

IN06 Preserve original and interpreted messages separately. AI classification is evidence, not an unreviewable overwrite of canonical Customer identity, verified requirements or existing ownership.

IN07 STOP, withdrawal of permission, wrong-person reports and human-takeover requests immediately stop inappropriate automation. Use deterministic checks before AI processing and human review for ambiguous language.

Successful progression

The campaign audience record remains linked after Lead creation. Materials requested, stated budget and meeting preference become attributed conversation evidence. The Lead remains governed by existing CORE assignment and compliance; neither campaign sending nor AI engagement grants access to restricted records.

Unexpected content

Unsupported documents, voice notes, images, suspicious links and languages outside the approved scope must remain visible for human handling. The first release must not silently discard them or claim that AI understood them. File-content interpretation is not included unless separately tested and approved.

10 AI follow through and human control

AI01 Enable AI separately from bulk dispatch, by approved campaign/channel and policy version. Provide a business-wide stop control and per-conversation human takeover. Bulk messaging must remain usable if AI is disabled.

AI02 The assistant may answer campaign-related questions, request missing requirements, share approved material, report authoritative price information and offer a callback or meeting. It must identify its assistant role and provide a clear human-contact path.

AI03 Use only permission-scoped CORE tools and approved material. No arbitrary SQL, unrestricted URLs, filesystem access, cross-customer search or unapproved external actions may be exposed to the model.

AI04 Treat uploads, brochures, customer messages and retrieved content as data, not tool instructions. Validate structured tool outputs, recipient identity, resource scope and action approval on the server before executing.

AI05 Do not promise availability, guaranteed investment return, finance approval, legal outcomes, discounts or contractual acceptance. Unknown or conflicting information must produce a truthful explanation and human handover.

AI06 Track buying/renting objective, business line, area, property type, fractional bedrooms, budget/currency, timeframe, purpose, language and meeting preference. Ask only useful missing questions and allow the prospect to decline.

AI07 Use the current customer-service window for free-form text/material responses; outside it, use an appropriate approved template and current permission. Re-evaluate the window at dispatch time, not when the job was scheduled. [S2]

AI08 Cancel pending AI replies and follow-ups when an Agent takes over or an opt-out occurs. A server-side generation check immediately before submission must defeat races with an already prepared AI draft.

Proposed operating bounds

Start with English and Hindi, a maximum 10 AI replies per conversation without meeting progress, and explicit token/message cost caps. Low confidence, repeated misunderstanding, complaint, unsupported language, sensitive decision or a direct request for a person creates a human task. These numerical/language defaults require approval in section 20. Human takeover cannot be automatically reversed by a later campaign.

11 Materials pricing and follow up

MAT01 Provide a versioned catalogue of approved prospectuses, brochures, floor plans, images, price sheets and marketing documents. Bind each item to its project/property, language, public-sharing permission, validity and approving actor.

MAT02 Only share a file that is approved, current and relevant to the request and campaign. Confirm configuration or project when ambiguous. Never expose title deeds, KYC, internal notes, owner contacts or private transaction documents as marketing material.

MAT03 Send through a supported provider document/media method or an approved recipient-appropriate link. Scan files, restrict MIME/size, prevent arbitrary URL fetches and private-network access, and validate provider media limits. Record exact file version and delivery attempt.

MAT04 Use CORE's authorised pricing source with currency, unit/configuration, basis, effective date and relevant qualifications. Distinguish starting price from a specific unit price. Recheck before sending; if a material price change undermines campaign approval, pause the affected sends for review.

MAT05 Never fabricate discounts, stock, payment plans, fees or price validity. A missing or stale price triggers a human task. Retain a snapshot of the exact price information supplied so later changes do not rewrite the conversation history.

FUP01 After a requested material is shared, allow only follow-ups specified in the approved package. Stop on reply where a newer intent supersedes the job, declined interest, opt-out, wrong number, handover, policy expiry or exhausted limits.

FUP02 Schedule in the agreed timezone/hours with a campaign frequency limit across lists. Proposal for approval: at most 2 unattended follow-ups, first after 24 hours and second after 72 hours, each subject to current window/template eligibility. These are proposed defaults, not existing agreed behaviour.

Example

A prospect asks for a two-bedroom floor plan and price. CORE selects the approved two-bedroom version and current pricing source. If several projects fit, it asks which project before sending. When the prospect requests a meeting, CORE cancels the material reminder and starts meeting/handover processing. If nothing is heard, only the approved bounded follow-up may run.

12 Meeting and accepted Agent handover

HND01 Use existing CORE team/business-line routing and ownership rules. Existing owned Leads should not be reassigned simply because they reply to a campaign. Where no eligible Agent is available, show a visible accountable queue and due task.

HND02 Offer a callback, office meeting, online meeting or viewing only within the approved capability. Real-time confirmed slots require an authorised connected calendar and an atomic reservation to prevent double booking.

HND03 Without calendar integration, collect preferred date, time and timezone, then label the request awaiting Agent confirmation. Do not send a confirmed invitation or describe a preferred time as booked.

HND04 Prepare a handover with campaign/recipient and Lead references, requirements, full conversation access, concise summary, exact material and pricing versions, outstanding questions, meeting status, next action and due time.

HND05 Agent acceptance must atomically record ownership/acceptance evidence and establish the next action. Notify the customer using an allowed message or template. Show a failed notification separately; it must not undo a valid acceptance or falsely claim delivery.

HND06 On acceptance, human ownership pauses autonomous sales responses. Agent messages must appear in the same authorised conversation. AI may resume only through an explicit allowed action; it cannot compete with a human for the next reply.

HND07 Escalate unaccepted or overdue handovers to the responsible Manager. Proposed acknowledgement target is 15 minutes during staffed hours, with the next-business-period fallback outside those hours. Business must approve hours, escalation recipient and timing.

Definitions for reporting

Outcome	Required evidence
Engaged	A relevant inbound response, not a sent or read callback.
Meeting requested	Prospect asks for a meeting or agrees to discuss preferred times.
Meeting confirmed	Actual confirmed slot and responsible person, not just a preference.
Handover accepted	Agent acceptance, owner and next action with due time.
Success case	Accepted handover, with meeting status stated accurately. A later completed meeting is a separate event.

13 User screens and responsibilities

Screen	Primary actions	Controls
WhatsApp setup	Admin enters protected sandbox/live credentials, tests channel and syncs templates.	No secret readback; distinct environments and sender bindings.
Recipient imports	Agent uploads, maps, reviews counts, corrects errors and names a batch.	Owner/team scope, row reasons, no silent partial success.
Audience segments	Agent filters, saves groups, previews eligibility and excludes recipients.	Unique-phone counts, explicit unknown values and permission reasons.
Template library	Draft wording, review internally and display provider status.	Version/language/category, last sync, approval history and blocked states.
Campaign composer	Choose audience, template, personalised content, assets, schedule and follow-through.	Exact preview; required fields; immutable submitted revision.
Manager review queue	Review template/campaign, approve, return or reject with reason.	Same actionable items as campaign details; no hidden pending approvals.
Campaign monitor	View totals, delivery, replies, costs, exclusions and unresolved attempts.	Card/detail/export reconciliation, pause, safe resume and audit history.
Conversation inbox	Review history, reply, share material, take over or hand over.	Ownership lock, current window, role scope and clear AI/human indicator.
Agent handover queue	Accept assignment, confirm meeting and complete next action.	Manager escalation, due times and existing CORE workflow links.

UI01 All cards, tables, exports and drill-downs use identical scope and count definitions. Pagination exposes the complete total; summaries never silently cap at 500.

UI02 Back and Close retain applied filters, selected page and scroll. Dialogs support keyboard access, clear return targets and responsive layouts. Action columns and document buttons must not collapse into individual letters.

UI03 Show a visible reason for blocked sends, failed saves and partial imports. Use plain names such as campaign approval or Agent handover; expose technical IDs only as secondary trace references.

14 Architecture and integration contracts

The browser calls authenticated CORE services. CORE stores campaign and conversation state, queues permitted actions and calls the provider through a bounded transport adapter. Incoming callbacks enter a durable inbox and are processed asynchronously. AI receives scoped facts and permitted tools; it does not call 360dialog or change CORE records independently.

Component	Responsibility
Identity and audience service	Validate imports, normalise channel identities, match CORE records, calculate segments and permission eligibility.
Template and approval service	Synchronise provider status, maintain internal versions and approve immutable campaign manifests.
Dispatch worker and outbox	Lease one action, recheck approvals/eligibility, reserve budget and submit through the correct environment.
Provider inbox	Authenticate, bind channel, deduplicate and persist encrypted inbound/status events before processing.
Conversation service	Resolve campaign context and identity, process opt-out/takeover, store history and control the messaging window.
AI orchestration	Interpret intent and prepare bounded actions using approved facts and deterministic server validation.
Material and handover services	Retrieve permitted versions, establish meeting status and create an accepted Agent next action.

Provider API boundaries

Sandbox uses only the fixed 360dialog sandbox host; live uses only the verified live host. User-supplied arbitrary API hosts and webhook URLs are prohibited. D360 API keys stay in server headers, never query strings or logs. Provider message IDs correlate callbacks; an API acceptance is not proof of delivery.

The template catalogue contract must prove account binding, pagination, component content, status names, notification subscription and access privileges on the real account. CORE must not advertise automatic verification complete until this contract test passes. The current live messages path and sandbox v1 paths are not interchangeable. [S1, S4]

Authentication and acknowledgement

Use authenticated HTTPS webhooks with replay/event-ID protection. The sandbox's custom authentication-header support must be demonstrated; if unavailable, resolve an approved authenticated relay contract before enabling receipt. Do not weaken the live receiver to accept unauthenticated events. Do not use public webhook-capture services for private messages. [S3]

15 Proposed data model and transaction rules

Entity	Principal data and constraints
WhatsApp connection	Environment, sender/account IDs, encrypted credentials, credential revision, callback configuration and health. Never return secrets.
Prospect and channel identity	Normalised phone or verified provider-scoped identity, optional profile, source, owner/team and resolved CORE links.
Import batch and row	File fingerprint, column mapping, row disposition, original evidence, validation reasons and batch reconciliation.
Permission and suppression	Identity/channel, purpose/scope, evidence, effective/withdrawal times and reason; append audit events.
Template version	Account/provider ID, name, locale, category, components, content fingerprint, provider status and observation time.
Campaign revision and recipient	Immutable approved manifest, recipient snapshot, rendered variables, material/pricing versions, schedule, budget and policy.
Approval event	Actor, permitted scope, decision, reason, approved revision hash, timestamp and expiry. Preserve previous decisions.
Attempt and provider event	Unique logical action, lease/generation, submit state, provider reference, delivery evidence, cost reservation and encrypted payload.
Conversation and follow-up	Participant/channel, history, last verified inbound time, linked campaigns/Lead, human owner, AI policy/generation and scheduled actions.
Material and handover link	Approved asset version and share permission; handover task, Agent acceptance, meeting status and next-action due time.

Proposed entity names are not final database DDL. Prefer explicit tables and foreign keys over opaque JSON for state, scope and uniqueness. Versioned manifests may use canonical JSON plus a stable hash. Monetary fields use decimal values and explicit currencies.

Database constraints enforce event deduplication, one logical send per revision/recipient/action, one promotion per conversation enquiry and atomic handover acceptance. Worker claims and suppression checks share a transactionally safe generation/lease contract. Status updates may arrive out of order; a late sent event must not downgrade delivered/read evidence. A restored queue must retain prior idempotency and provider references.

16 Security privacy and operational controls

SEC01 Admin alone manages secrets and sender setup. Agents access their permitted prospects and conversations; Manager scope follows existing managed teams; Director scope follows existing company access. Enforce scope on the server for reads, exports and writes.

SEC02 Encrypt credentials and sensitive message evidence using managed server keys, separate environments, private backups and audited rotation. Mask phone numbers in summaries where appropriate. Never place keys, KYC or private customer payloads in source, screenshots or diagnostic logs.

SEC03 Protect credential and campaign actions against cross-site submission, insecure direct object access, concurrent updates and stale revision approval. Rate-limit both public webhooks and authenticated administrative actions.

SEC04 Use approved AI data-processing arrangements, minimal required context and retention settings. Do not use customer conversation data for general model training. Business must approve the AI provider and information sent to it before activation. [S7]

OPS01 Monitor queue age, callback receipt and processing lag, template sync freshness, credential failures, send rejection, outcome unknown, budget use, opt-outs and handover age. An outage must be visible in the responsible operating queue.

OPS02 Keep an immutable audit trail for import correction, permission changes, approvals, campaign launch/pause, actual submitted content, material versions, price snapshots, AI actions and ownership transitions.

OPS03 Use a private recovery plan covering database, files, registered configuration, dependencies and encryption keys. Restore must preserve suppression and accepted sends; never restart dispatch automatically after restore without reconciliation.

Retention and capacity decisions

Business must set retention for source uploads, prospect profiles, message content, delivery evidence and audit records, with deletion/export responsibilities and permitted legal holds. Proposed starting points are 30 days for raw imports and 12 months for conversation/audit evidence, subject to review. Suppression identifiers may need longer controlled retention to prevent recontact; this is not a blanket deletion exemption.

Proposed first-release capacity is 5,000 imported rows per file and 1,000 eligible recipients per campaign, with asynchronous processing and configurable lower limits. These are design targets to be load-tested and approved, not current provider entitlement or an existing promise.

17 Sandbox and live channel acceptance

The sandbox recipient is the privately recorded test phone ending 6469. The API key is entered directly in a protected Admin setup form after that form is deployed and verified. It is not pasted into chat, uploaded in a spreadsheet or copied into source control. Saving it must not activate live outreach.

Test	Evidence required
SB01 Protected setup	Only Admin saves encrypted credentials; no API/UI/log readback. Test and live secrets are independent.
SB02 Recipient boundary	Only the approved recipient can receive a test send. A second phone and all production hosts are blocked.
SB03 Authenticated inbound	A real phone message reaches the private Test receiver; unauthenticated/wrong-channel events are rejected.
SB04 Replay and correlation	The same event retried creates one event/message; later replies use the same conversation.
SB05 Outbound text	An explicit test reply reaches the approved phone and receives a stored provider reference.
SB06 Template send	A currently available predefined sandbox template reaches the phone with correct variables.
SB07 Status callbacks	Sent/delivered/read or available failure events correlate to the correct attempt without status downgrade.
SB08 Failure control	Expired key, quota rejection, timeout, restart and pause produce visible states without blind retry or leakage.

Current provider documentation states 25 sandbox requests reaching the messaging API, including rejected messages; quota and template availability must be reconfirmed at execution. Earlier notes quoting 200 are superseded by the currently retrieved documentation. Use a conservative local test budget and avoid automated retry loops. The sandbox provides predefined templates and cannot prove real bulk throughput, custom-template approval, full media handling or live account limits. [S4]

After sandbox success, a separately approved live-channel canary must verify custom-template sync and status changes, approved document/image delivery, limited consenting recipients, real opt-out, frequency/budget controls, manual takeover and queue recovery. Sandbox success alone does not authorise REM removal or production activation.

18 Functional and regression acceptance

ID	Scenario and required result
AT01	Import 201 synthetic rows with mixed dispositions; exact reconciliation and row-level error export; no silent loss.
AT02	Reimport the same and overlapping files; no duplicate audience send; existing Customer identity/owner stays unchanged.
AT03	Import name and phone only; optional fields remain absent; non-customers do not become Customers/Leads automatically.
AT04	Saved segment changes after approval; approved recipients/messages stay fixed. New recipients require approval.
AT05	Missing or withdrawn permission and duplicate phone are excluded at preview and again at dispatch.
AT06	Agent cannot approve/send an unapproved campaign. Manager cannot approve another team outside scope. Self-approval follows the approved decision.
AT07	Message variables, materials, sender, schedule or recipients change; approval is invalidated as specified.
AT08	Pending/rejected/paused/deleted template or stale provider status blocks sending; a valid internal approval cannot bypass it.
AT09	Template language/components or content change externally; queued campaign pauses pending review.
AT10	Concurrent workers, duplicate launch and restart submit each logical action once; unknown outcomes are not resent blindly.
AT11	Out-of-order/repeated callbacks retain correct status and counts. Cards, details, pagination and exports reconcile beyond 500 rows.
AT12	New interested sender creates one governed enquiry; existing appropriate Lead is linked; shared-phone ambiguity goes to review.
AT13	STOP races with a prepared AI reply or leased follow-up; no subsequent unsubmitted marketing action is sent.
AT14	AI request for brochure/plan/price returns only approved, relevant current versions, or truthful human escalation.
AT15	Expired file, private KYC, invented discount, unsupported unit price and prompt-injection attempt cannot trigger disclosure or unauthorised action.

19 Handover tests and release criteria

ID	Scenario and required result
AT16	Customer-service window expires while a response waits; dispatch uses an eligible approved template or holds, never sends an ineligible free-form message.
AT17	Agent takes over while AI is preparing a reply; generation check cancels the AI action and the human owns the conversation.
AT18	No calendar connection; meeting preference is awaiting confirmation. No false confirmed booking or invite appears.
AT19	Two users select the same real slot or accept one handover concurrently; one valid reservation/acceptance results.
AT20	Accepted handover contains full history, requirements, exact shared assets, meeting status, owner, next action and due time.
AT21	Unaccepted handover exceeds the approved SLA; Manager sees an actionable escalation, not a hidden detail-only task.
AT22	Budget, frequency or quiet-hour limit is reached; sends defer/stop visibly without losing their original approval and count disposition.
AT23	Provider outage, credential rotation, unknown outcome and recovery restore preserve suppression, accepted attempts and deduplication.
AT24	Back/Close, applied filters, list page, scroll, modal focus and mobile action widths pass for Agent, Manager, Director and Admin.
AT25	Existing CRM regression, authorisation, all ten governed PDF formats, intake, tasks, dashboards and migration/rollback checks remain green.

Release evidence

Each requirement must map to implemented source, a repeatable test and a business acceptance record. Record skipped tests and explain whether they prevent release. Mark provider simulations, sandbox tests and live-channel evidence separately. Do not describe a mock response, saved key or accepted send as live end-to-end success.

Release only when blocking tests pass, no critical/high unresolved defect affects sending or customer data, the operating owner is trained and the cutover/recovery plan is approved. Package from an advertised GitHub commit through the mandatory origin builder and verify provenance before deployment. Verify the exact application-root workers before stop and exactly one worker after readiness.

Production activation needs exact-version, configuration, migration, spending and maintenance-window approval with fresh paired backup and verified encrypted off-server recovery. A schema rollback must preserve evidence and follow documented restore rules; destructive down-migration is not assumed safe.

20 Delivery plan decisions and signoff

Stage	Deliverable	Dependency
A Sandbox foundation	Protected key entry; authenticated receive; test text/template sends; status and replay evidence.	Sandbox key entered privately and approved recipient available.
B Bulk messaging	Import/segments; provider template sync; Agent preparation; Manager/Director approval; bounded dispatch and reporting.	Decisions below, approved live account/templates and test recipients.
C AI follow through	Approved materials/pricing, qualification, follow-ups, human takeover, meeting requests and accepted handover.	Approved AI provider/policy, catalogue, ownership and appointment capability.
D User release	Integrated UAT, recovery rehearsal, governed package and controlled canary.	Exact production approval. Target 8 November 2026; no automatic scope reduction or unsafe activation to meet it.

Stages B and C form the requested next user delivery. If dependencies threaten the target, present a dated exception and an explicit staged-release choice; do not silently release bulk-only while claiming AI complete.

Decisions required before the affected baseline

ID	Decision to approve or amend
D01	Independent reviewer required when a Manager/Director prepares a campaign; proposed yes.
D02	Approve proposed capacity 5,000 rows/file and 1,000 eligible recipients/campaign, or provide expected peak volume.
D03	Set sending hours, timezone fallback, cross-campaign frequency cap and maximum campaign spend. These must be explicit before live send.
D04	Approve/amend follow-ups: maximum 2 unattended attempts at 24 and 72 hours; approve wording and campaign-specific expiry.
D05	Approve AI provider, English/Hindi scope, 10-reply handover threshold, token/cost limits and data-processing terms.
D06	Name operating owner/team, staffed hours,15-minute handover target and escalation responsibility; preserve existing routing.
D07	Approve raw-import/conversation retention and deletion responsibilities, including controlled suppression retention.
D08	Choose initial appointment capability and approve catalogue/pricing owners, refresh intervals and template freshness default 5 minutes.

Approval record: business approver _____ decision _____ date/time _____

Scope and version approved _____ outstanding decisions/exclusions _____

Baseline only the approved document version and explicit decision resolutions. Record the final SHA256 and approving actor/date in the change register. Subsequent requirement changes require a new revision and impact review. Approval is not assumed from delivery of this draft.

21 Traceability sources and document control

Requirement group	Design coverage	Acceptance
WA01 and WA07	Sandbox boundary; staged bulk then AI delivery	SB01 to SB08 and AT25
WA02	Import, identity, audience and permission	AT01 to AT05 and AT12
WA03 and WA05	Template verification and immutable NYSA approvals	AT06 to AT11
WA04	AI, approved material, pricing and follow-up	AT13 to AT17 and AT22
WA08	Existing scope, routing, human takeover and release controls	AT18 to AT25
WA06	Fresh templates; REM cutover remains separate	Template/live canary and cutover review

Primary references

Provider references were checked on 1 November 2026. They establish transport constraints, not the implementation or approval status of CORE. Recheck account-specific capabilities before activation. Numerical operational defaults in this document are proposals, not provider limits unless expressly attributed.

S1 360dialog Templates
https://docs.360dialog.com/docs/resources/templates

S2 360dialog Free versus billed messaging
https://docs.360dialog.com/docs/get-started/pricing/free-vs-billed-messaging

S3 360dialog Webhooks
https://docs.360dialog.com/docs/messaging/webhook

S4 360dialog Sandbox current documentation
https://docs.360dialog.com/docs/get-started/sandbox

S5 360dialog Marketing messages
https://docs.360dialog.com/docs/resources/marketing-messages

S6 360dialog Migrations
https://docs.360dialog.com/docs/hub/migrations

S7 WhatsApp Business Solution Terms
https://www.whatsapp.com/legal/business-solution-terms

S8 360dialog Opt in
https://docs.360dialog.com/docs/messaging/overview/opt-in

Internal evidence and change boundary

Baseline evidence: BASELINE.md and release-artifacts/dev254-deployment. Inspected transport: src/lead-whatsapp.js; inbox: src/routes/lead-channel-inbox.js; protected connections: src/lead-integration-connections.js; setup: src/routes/lead-integration-setup.js. The last observed Test/production health responses were DEV254/RC1. This document makes no runtime, database, provider or REM change.